TSP Optimization with Simulated Annealing and Genetic Algorithms — Assignment Notebook

Works in **Jupyter** and **Google Colab**.

Overview

You will solve the same TSP instance using **two heuristic optimization methods**:

1. **Simulated Annealing (SA)** with a 2-opt neighborhood
2. **Genetic Algorithm (GA)** on **permutation tours** (a standard evolutionary approach for TSP)

The notebook provides:

- A reproducible TSP instance generator
- Tour utilities (length, validity)
- A 2-opt neighborhood operator
- Visualization helpers (tour plot + history plot)
- Baseline heuristic: nearest neighbor

Tasks

Task A — Simulated Annealing (SA)

Implement `simulated_annealing_tsp(...)`.

Task B — Genetic Algorithm (GA)

Implement `genetic_algorithm_tsp(...)` plus helper functions:

- Tournament selection
- Order crossover (OX)
- Swap mutation

Deliverables

- Completed notebook (or exported `.py`) (find all TODOs and implement them)
- Short write-up (more detail in Part 6):
  - SA parameters + best length

- GA parameters + best length
- Brief comparison: which worked better and why

In [1]:
```
# If running in a fresh environment, uncomment as needed:
# !pip -q install matplotlib

import math
import random
from dataclasses import dataclass
from typing import List, Tuple

import matplotlib.pyplot as plt
```

Part 1 — TSP utilities (provided)

In [2]:
```
Point = Tuple[float, float]
Tour = List[int]

def make_cities(n: int = 40, seed: int = 7) -> List[Point]:
    # Reproducible 2D city coordinates in [0, 1]x[0, 1].
    rng = random.Random(seed)
    return [(rng.random(), rng.random()) for _ in range(n)]

def dist(a: Point, b: Point) -> float:
    dx = a[0] - b[0]
    dy = a[1] - b[1]
    return math.hypot(dx, dy)

def tour_length(cities: List[Point], tour: Tour) -> float:
    # Total length of a closed tour.
    n = len(tour)
    total = 0.0
    for i in range(n):
        total += dist(cities[tour[i]], cities[tour[(i + 1) % n]])
    return total

def is_valid_tour(tour: Tour, n: int) -> bool:
    return len(tour) == n and set(tour) == set(range(n))

def random_tour(n: int, rng: random.Random) -> Tour:
    t = list(range(n))
    rng.shuffle(t)
    return t

def nearest_neighbor_tour(cities: List[Point], start: int = 0) -> Tour:
    # Deterministic baseline: greedy nearest neighbor.
    n = len(cities)
    unvisited = set(range(n))
    tour = [start]
    unvisited.remove(start)
    cur = start
    while unvisited:
```

```
        nxt = min(unvisited, key=lambda j: dist(cities[cur], cities[j]))
        tour.append(nxt)
        unvisited.remove(nxt)
        cur = nxt
    return tour
```

Part 2 — Neighborhood operator (2-opt) (provided)

In [3]:
```
def two_opt_swap(tour: Tour, i: int, k: int) -> Tour:
    # Return a new tour where the segment [i:k] is reversed.
    return tour[:i] + list(reversed(tour[i:k + 1])) + tour[k + 1:]

def random_two_opt_neighbor(tour: Tour, rng: random.Random) -> Tour:
    # Pick a random 2-opt move.
    n = len(tour)
    i = rng.randrange(0, n - 1)
    k = rng.randrange(i + 1, n)
    return two_opt_swap(tour, i, k)
```

Part 3 — Visualization helpers (provided)

- You are encouraged to implement more variations of visualization.

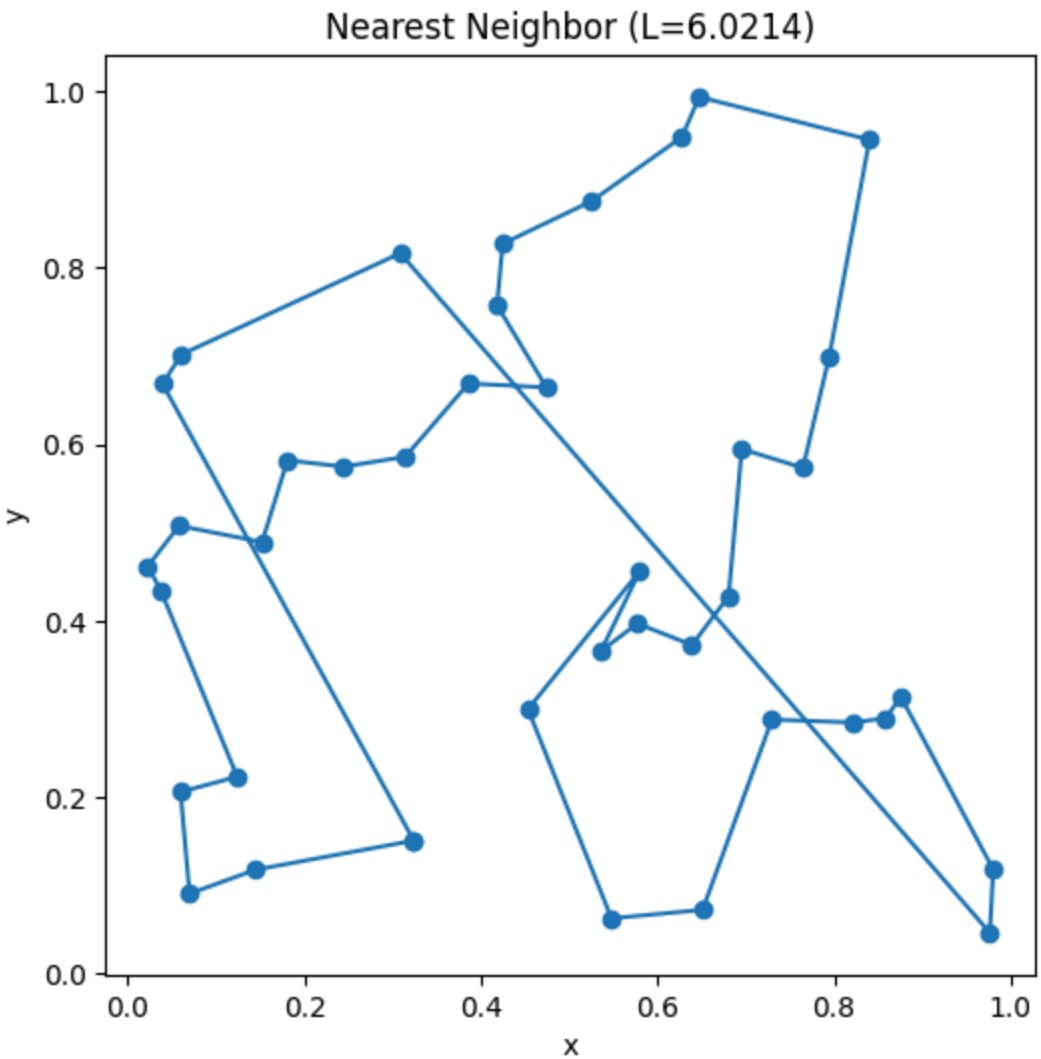
In [4]:
```
def plot_tour(cities: List[Point], tour: Tour, title: str = "") -> None:
    xs = [cities[i][0] for i in tour] + [cities[tour[0]][0]]
    ys = [cities[i][1] for i in tour] + [cities[tour[0]][1]]

    plt.figure(figsize=(6, 6))
    plt.plot(xs, ys, marker="o")
    plt.title(title)
    plt.xlabel("x")
    plt.ylabel("y")
    plt.gca().set_aspect("equal", adjustable="box")
    plt.show()

def plot_history(series: List[float], title: str, ylabel: str = "Value") ->
    plt.figure(figsize=(7, 4))
    plt.plot(series)
    plt.title(title)
    plt.xlabel("Iteration")
    plt.ylabel(ylabel)
    plt.show()

def plot_compare(sa_hist: List[float], ga_hist: List[float]) -> None:
    # Overlay histories (x-axis is not identical meaning; this is a visual c
    plt.figure(figsize=(8, 4))
    plt.plot(sa_hist, label="SA best-so-far")
    plt.plot([i * (len(sa_hist) / max(1, len(ga_hist)-1)) for i in range(ler
    plt.title("SA vs GA: Best tour length over time (scaled x-axis)")
    plt.xlabel("Progress (scaled)")
    plt.ylabel("Length")
```

```
    plt.legend()
    plt.show()
```

Part 4 — Baseline tour (nearest neighbor) (provided)

In [5]:
```
cities = make_cities(n=40, seed=7)

nn_tour = nearest_neighbor_tour(cities, start=0)
nn_len = tour_length(cities, nn_tour)
print(f"Nearest-neighbor length: {nn_len:.4f}")

plot_tour(cities, nn_tour, title=f"Nearest Neighbor (L={nn_len:.4f})")
```

```
Nearest-neighbor length: 6.0214
```



Part 5A — Code Skeleton: Simulated Annealing (SA)

Complete `simulated_annealing_tsp(...)`.

SA acceptance rule

Let `delta = cand_len - cur_len`.

- If `delta <= 0`, accept.
- Else accept with probability `exp(-delta / T)`.

Cooling schedule

Geometric cooling:

- `T = T * alpha` each iteration
- clamp `T = max(T, 1e-12)`

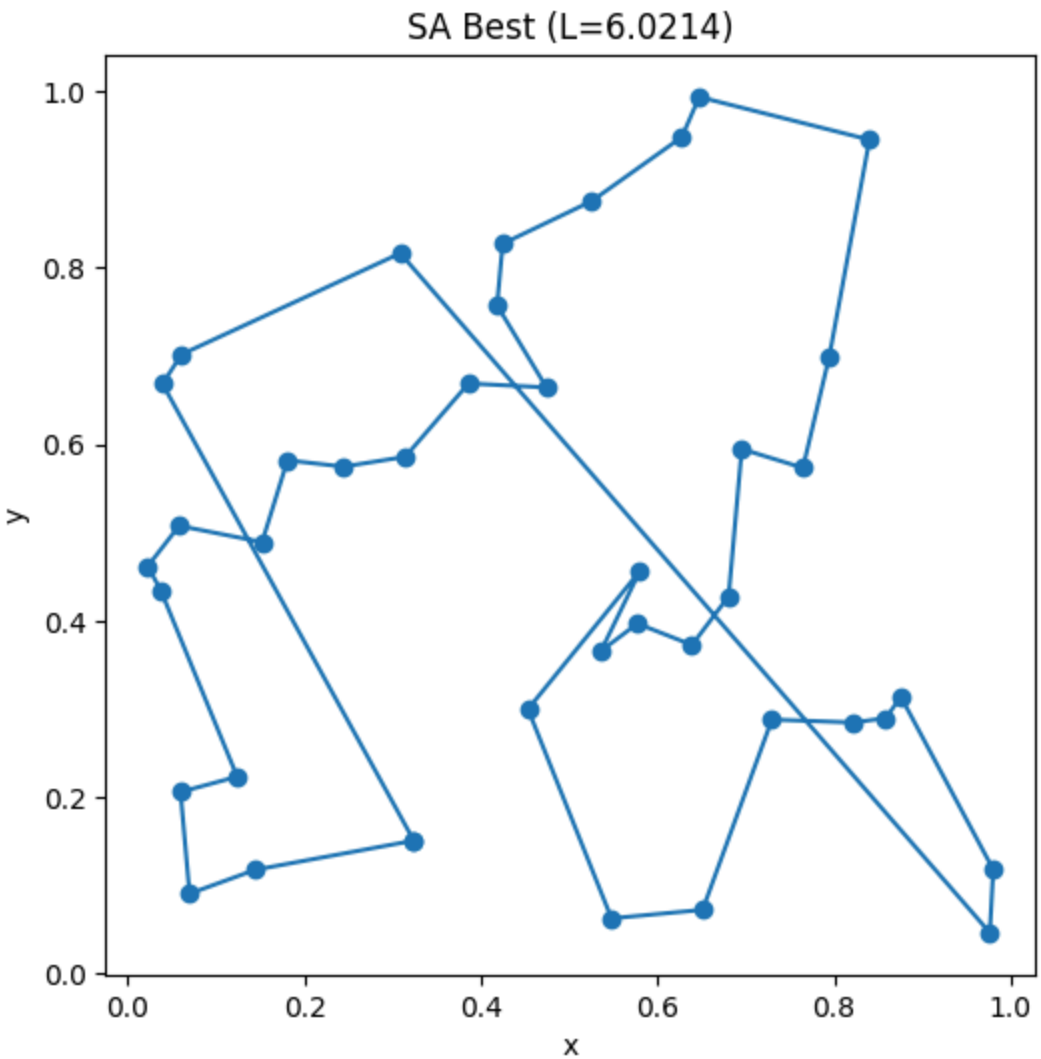
In [6]:
```
@dataclass
class SAConfig:
    iters: int = 20_000
    t0: float = 0.2
    alpha: float = 0.9995
    seed: int = 123
    report_every: int = 2000

def simulated_annealing_tsp(cities: List[Point], init_tour: Tour, cfg: SACor
    rng = random.Random(cfg.seed)
    n = len(cities)
    assert is_valid_tour(init_tour, n), "init_tour must be a valid permutati

    cur_tour = init_tour[:]
    cur_len = tour_length(cities, cur_tour)

    best_tour = cur_tour[:]
    best_len = cur_len
    history = [best_len]

    T = cfg.t0

    for it in range(cfg.iters):
        cand_tour = random_two_opt_neighbor(cur_tour, rng)
        cand_len = tour_length(cities, cand_tour)
        delta = cand_len - cur_len

        accept = False
        # TODO: acceptance rule
        # Is accept True or false when delta <= 0 or delta >0?

        if accept:
            cur_tour, cur_len = cand_tour, cand_len

        if cur_len < best_len:
            best_tour, best_len = cur_tour[:], cur_len

        history.append(best_len)
```

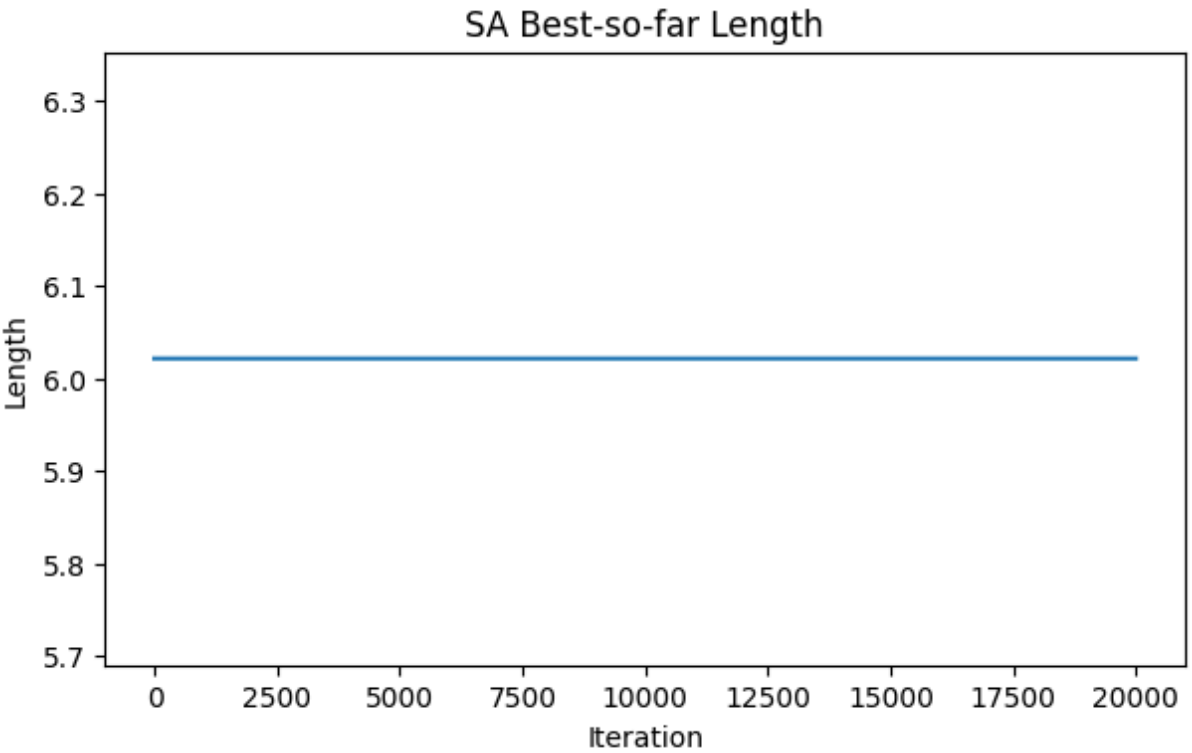
```
        # TODO: cooling schedule
        # How is T adjusted?

        if cfg.report_every and (it + 1) % cfg.report_every == 0:
            print(f"[SA] iter={it+1:6d}  T={T:.4g}  cur={cur_len:.4f}  best=

    return best_tour, best_len, history
```

Run SA (after you implement it)

In [7]:
```
sa_cfg = SAConfig(iters=20_000, t0=0.2, alpha=0.9995, seed=123, report_every
sa_best_tour, sa_best_len, sa_hist = simulated_annealing_tsp(cities, nn_tour

print(f"SA best length: {sa_best_len:.4f}")
print(f"SA improvement vs NN: {(nn_len - sa_best_len) / nn_len * 100:.2f}%")

assert is_valid_tour(sa_best_tour, len(cities))

plot_tour(cities, sa_best_tour, title=f"SA Best (L={sa_best_len:.4f})")
plot_history(sa_hist, title="SA Best-so-far Length", ylabel="Length")
```

```
[SA] iter=  2000  T=0.2  cur=6.0214  best=6.0214
[SA] iter=  4000  T=0.2  cur=6.0214  best=6.0214
[SA] iter=  6000  T=0.2  cur=6.0214  best=6.0214
[SA] iter=  8000  T=0.2  cur=6.0214  best=6.0214
[SA] iter= 10000  T=0.2  cur=6.0214  best=6.0214
[SA] iter= 12000  T=0.2  cur=6.0214  best=6.0214
[SA] iter= 14000  T=0.2  cur=6.0214  best=6.0214
[SA] iter= 16000  T=0.2  cur=6.0214  best=6.0214
[SA] iter= 18000  T=0.2  cur=6.0214  best=6.0214
[SA] iter= 20000  T=0.2  cur=6.0214  best=6.0214
SA best length: 6.0214
SA improvement vs NN: 0.00%
```


SA Best (L=6.0214)
1.0
0.8
0.6
0.4
0.2
0.0
y
0.0
0.2
0.4
0.6
0.8
1.0
x


SA Best-so-far Length
6.3
6.2
6.1
6.0
5.9
5.8
5.7
Length
0
2500
5000
7500
10000
12500
15000
17500
20000
Iteration

Part 5B — Code Skeleton: Genetic Algorithm (GA) for TSP

Implement a GA that evolves a population of permutation tours.

You will implement:

- `tournament_select(...)`
- `order_crossover_ox(...)` (Order Crossover)
- `mutate_swap(...)`
- `genetic_algorithm_tsp(...)`

In [8]:
```
@dataclass
class GAConfig:
    pop_size: int = 200
    generations: int = 400
    tournament_k: int = 5
    crossover_rate: float = 0.9
    mutation_rate: float = 0.2
    elite_size: int = 5
    seed: int = 999
    report_every: int = 50

def tournament_select(pop: List[Tour], lengths: List[float], k: int, rng: ra
    # TODO: tournament selection (minimize length)
    k = min(k, len(pop))
    cand_idxs = rng.sample(range(len(pop)), k)
    best_i = min(cand_idxs, key=lambda i: lengths[i])  # shortest wins
    return pop[best_i][:]  # copy
```

Run GA

- After you implement TODOs above
- This is just for your reference, you can experiment with different paramesters

In [9]:
```
from dataclasses import dataclass
from typing import List, Tuple
import random

# Safe aliases in case they aren't defined yet (won't hurt if they already a
Point = Tuple[float, float]
Tour = List[int]

@dataclass
class GAConfig:
    pop_size: int = 200
    generations: int = 400
    tournament_k: int = 5
    crossover_rate: float = 0.9
```

```
    mutation_rate: float = 0.2
    elite_size: int = 5
    seed: int = 999
    report_every: int = 50

def tournament_select(pop: List[Tour], lengths: List[float], k: int, rng: ra
    k = min(k, len(pop))
    cand_idxs = rng.sample(range(len(pop)), k)
    best_i = min(cand_idxs, key=lambda i: lengths[i])  # minimize length
    return pop[best_i][:]

def order_crossover_ox(parent1: Tour, parent2: Tour, rng: random.Random) ->
    n = len(parent1)
    a, b = sorted(rng.sample(range(n), 2))
    child = [None] * n

    # copy slice from parent1
    child[a:b+1] = parent1[a:b+1]
    used = set(child[a:b+1])

    # fill from parent2 starting at b+1 (wrap), skipping used
    fill = []
    for gene in (parent2[b+1:] + parent2[:b+1]):
        if gene not in used:
            fill.append(gene)

    # place into empty spots starting at b+1 (wrap)
    pos_order = list(range(b+1, n)) + list(range(0, b+1))
    j = 0
    for pos in pos_order:
        if child[pos] is None:
            child[pos] = fill[j]
            j += 1

    return child  # type: ignore

def mutate_swap(tour: Tour, rng: random.Random) -> Tour:
    i, j = rng.sample(range(len(tour)), 2)
    out = tour[:]
    out[i], out[j] = out[j], out[i]
    return out

def genetic_algorithm_tsp(cities: List[Point], init_seed_tours: List[Tour],
    rng = random.Random(cfg.seed)
    n = len(cities)

    pop: List[Tour] = []
    for t in init_seed_tours:
        assert is_valid_tour(t, n)
        pop.append(t[:])
    while len(pop) < cfg.pop_size:
        pop.append(random_tour(n, rng))

    lengths = [tour_length(cities, t) for t in pop]
    best_idx = min(range(len(pop)), key=lambda i: lengths[i])
    best_tour = pop[best_idx][:]
```

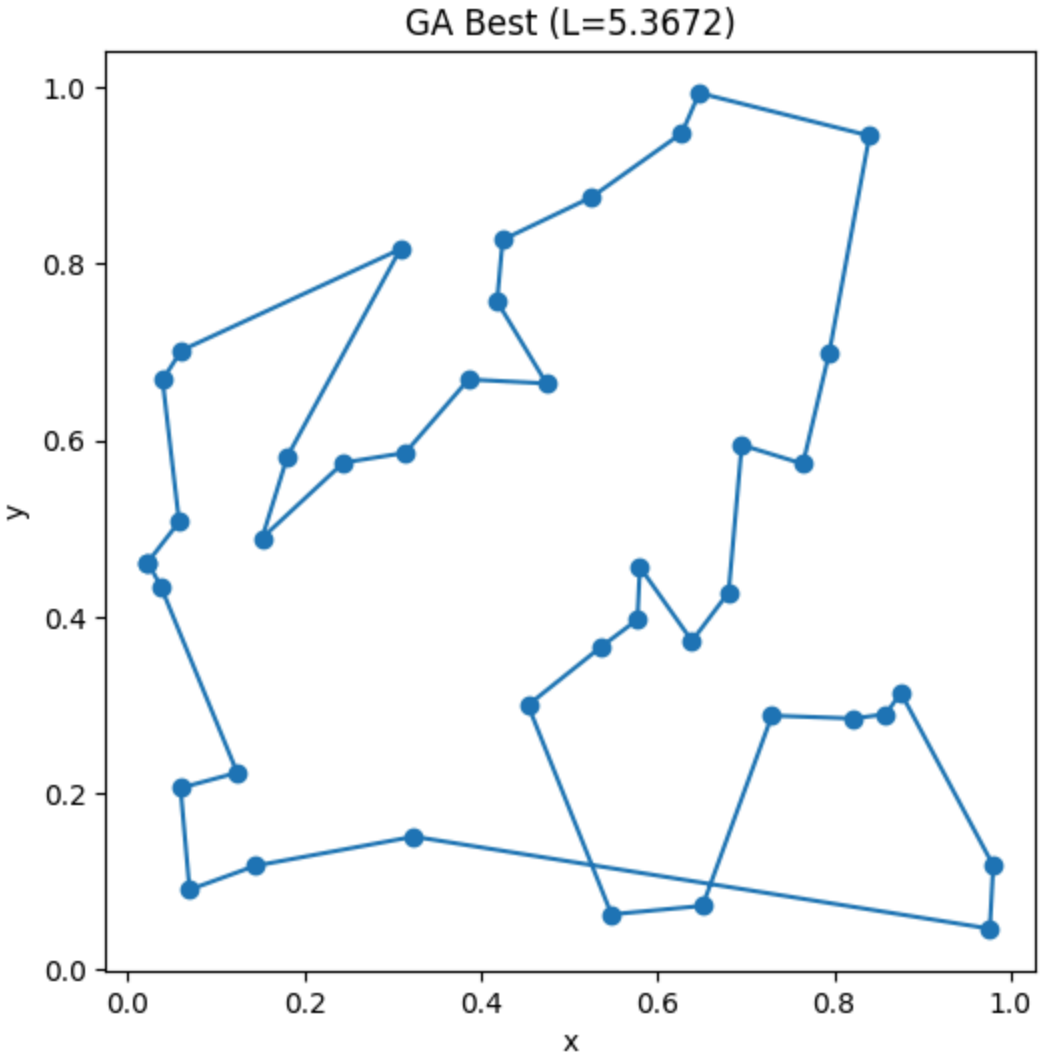
```
    best_len = lengths[best_idx]
    history = [best_len]

    for gen in range(cfg.generations):
        elite_indices = sorted(range(len(pop)), key=lambda i: lengths[i])[:
        next_pop = [pop[i][:] for i in elite_indices]

        while len(next_pop) < cfg.pop_size:
            p1 = tournament_select(pop, lengths, cfg.tournament_k, rng)
            p2 = tournament_select(pop, lengths, cfg.tournament_k, rng)

            if rng.random() < cfg.crossover_rate:
                child = order_crossover_ox(p1, p2, rng)
            else:
                child = p1[:]

            if rng.random() < cfg.mutation_rate:
                child = mutate_swap(child, rng)

            next_pop.append(child)

        pop = next_pop
        lengths = [tour_length(cities, t) for t in pop]

        gen_best_idx = min(range(len(pop)), key=lambda i: lengths[i])
        gen_best_len = lengths[gen_best_idx]
        if gen_best_len < best_len:
            best_len = gen_best_len
            best_tour = pop[gen_best_idx][:]

        history.append(best_len)

        if cfg.report_every and (gen + 1) % cfg.report_every == 0:
            print(f"[GA] gen={gen+1:4d}  best={best_len:.4f}")

    assert is_valid_tour(best_tour, n)
    return best_tour, best_len, history
```

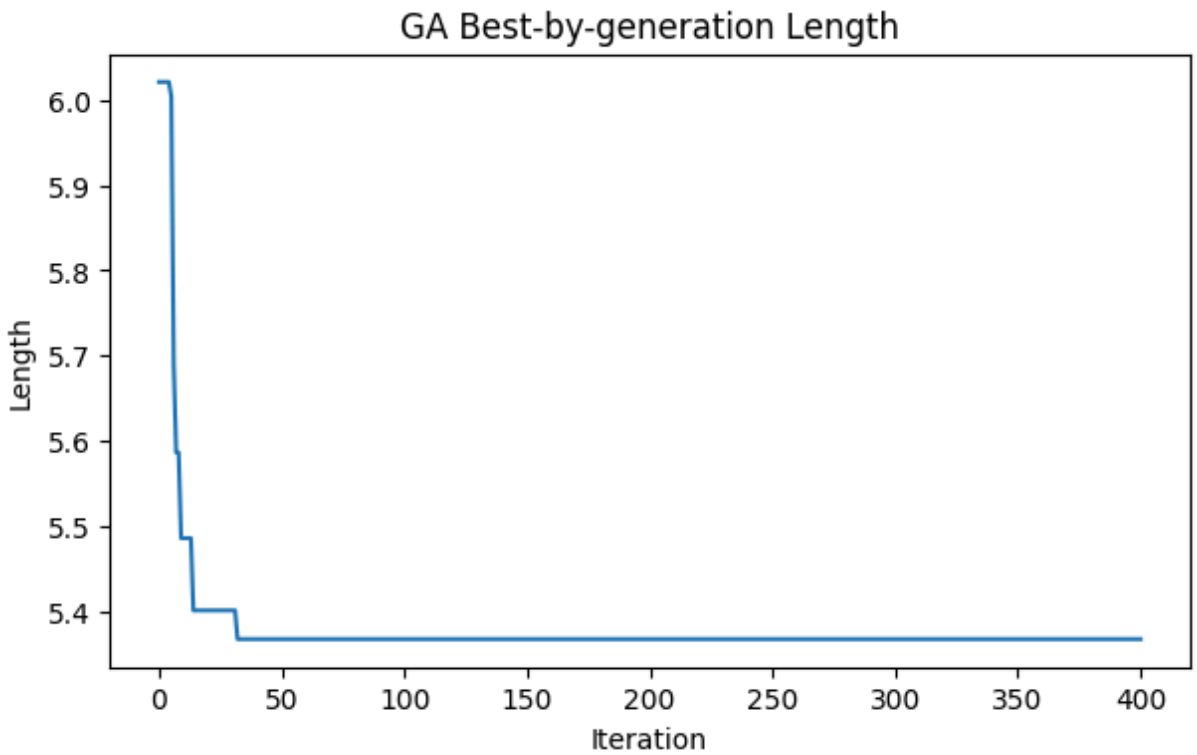
In [10]:

```
ga_cfg = GAConfig(pop_size=200, generations=400, tournament_k=5, crossover_r
                  mutation_rate=0.2, elite_size=5, seed=999, report_every=5

ga_best_tour, ga_best_len, ga_hist = genetic_algorithm_tsp(cities, init_seed

print(f"GA best length: {ga_best_len:.4f}")
print(f"GA improvement vs NN: {(nn_len - ga_best_len) / nn_len * 100:.2f}%")

plot_tour(cities, ga_best_tour, title=f"GA Best (L={ga_best_len:.4f})")
plot_history(ga_hist, title="GA Best-by-generation Length", ylabel="Length")
```

```
[GA] gen=  50  best=5.3672
[GA] gen= 100  best=5.3672
[GA] gen= 150  best=5.3672
[GA] gen= 200  best=5.3672
[GA] gen= 250  best=5.3672
[GA] gen= 300  best=5.3672
[GA] gen= 350  best=5.3672
```

```
[GA] gen= 400  best=5.3672
GA best length: 5.3672
GA improvement vs NN: 10.86%
```





Part 6 — Data Collection, Visualization and Comparison

- Instrument the code to collecte the data and visualize them, including but not limited to
  - SA:
    - Track histogram of ΔE values for: Accepted moves and Rejected moves
  - GA:
    - Population average fitness
  - Cross SA and GA:
    - Plot SA best-so-far curve
    - Plot GA best-by-generation curve
    - Normalize x-axis by total evaluations
    - The above is already provided in the skeleton code. **Use them and/or with additional instrumentation of the code to collect and/or plot more data** to answer the following questions:
      - Which algorithm converged faster?
      - Which showed more variability?
      - Did SA escape local minima? Evidence?
      - Did GA lose diversity? Evidence?
- You are encouraged to compare more extensively.

In [11]:
```
# After both SA and GA work, run:
# plot_compare(sa_hist, ga_hist)

# And print a quick summary:
# print(f"NN: {nn_len:.4f} | SA: {sa_best_len:.4f} | GA: {ga_best_len:.4f}")

# TODO
```

Example of how to check your work

In [12]:
```
def grade_check(nn_len: float, method_len: float, min_improvement_pct: float
    improvement = (nn_len - method_len) / nn_len * 100.0
    print(f"{label} length: {method_len:.4f} | Improvement vs NN: {improveme
    assert improvement >= min_improvement_pct, (
        f"{label} improvement {improvement:.2f}% is below required {min_impr
    )
    print(f"✅ {label} passed improvement threshold!")
```

Part 7: Optional Task

Try different crossover and/or mutation operations in GA and see if you can improve the performance. Document your effort in this section.